A2.8 Return Decomposition to construct a Reward Redistribution

We now propose methods to construct reward redistributions which ideally would be optimal. Learning with non-optimal reward redistributions *does work* since the optimal policies do not change according to Theorem A2. However reward redistributions that are optimal considerably speed up learning, since future expected rewards introduce biases in TD-methods and the high variance in MC-methods. The expected optimal redistributed reward is according to Eq. (A65) the difference of Q-values. The more a reward redistribution deviates from these differences, the larger are the absolute κ-values and, in turn, the less optimal is the reward redistribution. Consequently we aim at identifying the largest Q-value differences to construct a reward redistribution which is close to optimal. Assume a grid world where you have to take a key to later open a door to a treasure room. Taking the key increases the chances to receive the treasure and, therefore, is associated with a large positive Q-value difference. Smaller positive Q-value difference are steps toward the key location.

Reinforcement Learning as Pattern Recognition. We want to transform the reinforcement learning problem into a pattern recognition problem to employ deep learning approaches. The sum of the Q-value differences gives the difference between expected return at sequence begin and the expected return at sequence end (telescope sum). Thus, Q-value differences allow to predict the expected return of the whole state-action sequence. Identifying the largest Q-value differences reduce the prediction error most. Q-value differences are assumed to be associated with patterns in state-action transitions like taking the key in our example. The largest Q-value differences are expected to be found more frequently in sequences with very large or very low return. The resulting task is to predict the expected return from the whole sequence and identify which state-action transitions contributed most to the prediction. This pattern recognition task is utilized to construct a reward redistribution, where redistributed reward corresponds to the contribution.

A2.8.1 Return Decomposition Idea

The *return decomposition idea* is to predict the realization of the return or its expectation by a function g from the state-action sequence

$$(s, a)_{0:T} \;:=\; (s_0, a_0, s_1, a_1, \ldots, s_T, a_T)\,. \tag{A119}$$

The return is the accumulated reward along the whole sequence $(s, a)_{0:T}$. The function g depends on the policy π that is used to generate the state-action sequences. Subsequently, the prediction or the realization of the return is distributed over the sequence with the help of g. One important advantage of a deterministic function g is that it predicts with proper loss functions and if being perfect the expected return. Therefore, it removes the sampling variance of returns. In particular the variance of probabilistic rewards is averaged out. Even an imperfect function g removes the variance as it is deterministic. As described later, the sampling variance may be reintroduced when strictly return-equivalent SDPs are ensured. We want to determine for each sequence element its contribution to the prediction of the function g. Contribution analysis computes the contribution of each state-action pair to the prediction, that is, the information of each state-action pair about the prediction. In principle, we can use any contribution analysis method. However, we prefer three methods: (A) Differences in predictions. If we can ensure that g predicts the sequence-wide return at every time step. The difference of two consecutive predictions is a measure of the contribution of the current state-action pair to the return prediction. The difference of consecutive predictions is the redistributed reward. (B) Integrated gradients (IG) [125]. (C) Layer-wise relevance propagation (LRP) [3]. The methods (B) and (C) use information later in the sequence for determining the contribution of the current state-action pair. Therefore, they introduce a non-Markov reward. However, the non-Markov reward can be viewed as probabilistic reward. Since probabilistic reward increases the variance, we prefer method (A).

Explaining Away Problem. We still have to tackle the problem that reward causing actions do not receive redistributed rewards since they are explained away by later states. To describe the problem, assume an MDP $\tilde{\mathcal{P}}$ with the only reward at sequence end. To ensure the Markov property, states in $\tilde{\mathcal{P}}$ have to store the reward contributions of previous state-actions; e.g. s_T has to store all previous contributions such that the expectation $\tilde{r}(s_T, a_T)$ is Markov. The explaining away problem is that later states are used for return prediction, while reward causing earlier actions are missed. To avoid explaining away, between the state-action pair (s_t, a_t) and its predecessor (s_{t-1}, a_{t-1}), where (s_{-1}, a_{-1}) are introduced for starting an episode. The sequence of differences is defined as

$$\Delta_{0:T} \;:=\; \big(\Delta(s_{-1}, a_{-1}, s_0, a_0), \ldots, \Delta(s_{T-1}, a_{T-1}, s_T, a_T)\big)\,. \tag{A120}$$

We assume that the differences Δ are mutually independent [60]:

$$p\left(\Delta(s_{t-1}, a_{t-1}, s_t, a_t) \mid \Delta(s_{-1}, a_{-1}, s_0, a_0), \ldots, \Delta(s_{t-2}, a_{t-2}, s_{t-1}, a_{t-1}),\right. \\ \left.\Delta(s_t, a_t, s_{t+1}, a_{t+1}) \ldots, \Delta(s_{T-1}, a_{T-1}, s_T, a_T)\right) \;=\; p\left(\Delta(s_{t-1}, a_{t-1}, s_t, a_t)\right) \;. \tag{A121}$$

The function g predicts the realization of the sequence-wide return or its expectation from the sequence $\Delta_{0:T}$:

$$g\big(\Delta_{0:T}\big) \;=\; \mathrm{E}\left[\tilde{R}_{T+1} \mid s_T, a_T\right] \;=\; \tilde{r}_{T+1} \;. \tag{A122}$$

Return decomposition deconstructs g into contributions $h_t = h(\Delta(s_{t-1}, a_{t-1}, s_t, a_t)$ at time t:

$$g\big(\Delta_{0:T}\big) \;=\; \sum_{t=0}^{T} h(\Delta(s_{t-1}, a_{t-1}, s_t, a_t)) \;=\; \tilde{r}_{T+1} \;. \tag{A123}$$

If we can assume that g can predict the return at every time step:

$$g\big(\Delta_{0:t}\big) \;=\; \mathrm{E}_\pi\left[\tilde{R}_{T+1} \mid s_t, a_t\right] \;, \tag{A124}$$

then we use the contribution analysis method "differences of return predictions", where the contributions are defined as:

$$h_0 \;=\; h(\Delta(s_{-1}, a_{-1}, s_0, a_0)) \;:=\; g\big(\Delta_{0:0}\big) \tag{A125}$$

$$h_t \;=\; h(\Delta(s_{t-1}, a_{t-1}, s_t, a_t)) \;:=\; g\big(\Delta_{0:t}\big) \;-\; g\big(\Delta_{0:(t-1)}\big) \;. \tag{A126}$$

We assume that the sequence-wide return cannot be predicted from the last state. The reason is that either immediate rewards are given only at sequence end without storing them in the states or information is removed from the states. Therefore, a relevant event for predicting the final reward must be identified by the function g. The prediction errors at the end of the episode become, in general, smaller since the future is less random. Therefore, prediction errors later in the episode are up-weighted while early predictions ensure that information is captured in h_t for being used later. The prediction at time T has the largest weight and relies on information from the past.
If g does predict the return at every time step, contribution analysis decomposes g. For decomposing a linear g one can use the Taylor decomposition (a linear approximation) of g with respect to the h [3, 83]. A non-linear g can be decomposed by layerwise relevance propagation (LRP) [3, 84] or integrated gradients (IG) [125].

A2.8.2 Reward Redistribution based on Return Decomposition

We assume a return decomposition

$$g\big(\Delta_{0:T}\big) \;=\; \sum_{t=0}^{T} h_t \;, \tag{A127}$$

with

$$h_0 \;=\; h(\Delta(s_{-1}, a_{-1}, s_0, a_0)) \;, \tag{A128}$$

$$h_t \;=\; h(\Delta(s_{t-1}, a_{t-1}, s_t, a_t)) \quad \text{for } 0 < t \leqslant T \;. \tag{A129}$$

We use these contributions for redistributing the reward. The reward redistribution is given by the random variable R_{t+1} for the reward at time $t+1$. These new redistributed rewards R_{t+1} must have the contributions h_t as mean:

$$\mathrm{E}\left[R_{t+1} \mid s_{t-1}, a_{t-1}, s_t, a_t\right] \;=\; h_t \tag{A130}$$

The reward $\tilde{R}_{T+1}$ of $\tilde{\mathcal{P}}$ is probabilistic and the function g might not be perfect, therefore neither $g(\Delta_{0:T}) = \tilde{r}_{T+1}$ for the return realization $\tilde{r}_{T+1}$ nor $g(\Delta_{0:T}) = \tilde{r}(s_T, a_T)$ for the expected return holds. To assure strictly return-equivalent SDPs, we have to compensate for both a probabilistic reward $\tilde{R}_{T+1}$ and an imperfect function g. The compensation is given by

$$\tilde{r}_{T+1} \;-\; \sum_{\tau=0}^{T} h_t \;. \tag{A131}$$

We compensate with an extra reward R_{T+2} at time $T+2$ which is immediately given after R_{T+1} at time $T+1$ after the state-action pair (s_T, a_T). The new redistributed reward R_{t+1} is

$$\mathrm{E}\left[R_1 \mid s_0, a_0\right] \;=\; h_0 \;, \tag{A132}$$

$$\mathrm{E}\left[R_{t+1} \mid s_{t-1}, a_{t-1}, s_t, a_t\right] \;=\; h_t \quad \text{for } 0 < t \leqslant T \;, \tag{A133}$$

$$R_{T+2} \;=\; \tilde{R}_{T+1} \;-\; \sum_{t=0}^{T} h_t \;, \tag{A134}$$

where the realization $\tilde{r}_{T+1}$ is replaced by its random variable $\tilde{R}_{T+1}$. If the the prediction of g is perfect, then we can set $R_{T+2} = 0$ and redistribute the expected return which is the predicted return. R_{T+2} compensates for both a probabilistic reward $\tilde{R}_{T+1}$ and an imperfect function g. Consequently all variance of sampling the return is moved to R_{T+2}. Only the imperfect function g must be corrected while the variance does not matter. However, we cannot distinguish, e.g. in early learning phases, between errors of g and random reward. **A perfect g results in an optimal reward redistribution.** Next theorem shows that Theorem A4 holds also for the correction R_{T+2}.

Theorem A6. *The optimality conditions hold also for reward redistributions with corrections:*

$$\kappa(T-t+1, t-1) \;=\; 0 \;. \tag{A135}$$

Proof. The expectation of $\kappa(T-t+1, t-1) = \sum_{\tau=0}^{T-t+1} R_{t+1+\tau}$, that is $\kappa(m, t-1)$ with $m = T-t+1$.

$$\begin{aligned}
&\mathrm{E}_\pi\left[\sum_{\tau=0}^{T-t+1} R_{t+1+\tau} \mid s_{t-1}, a_{t-1}\right] && \text{(A136)}\\
&= \mathrm{E}_\pi\left[\tilde{R}_{T+1} \;-\; \tilde{q}^\pi(s_T, a_T) \;+\; \sum_{\tau=0}^{T-t} \left(\tilde{q}^\pi(s_{\tau+t}, a_{\tau+t}) \;-\; \tilde{q}^\pi(s_{\tau+t-1}, a_{\tau+t-1})\right) \mid s_{t-1}, a_{t-1}\right]\\
&= \mathrm{E}_\pi\left[\tilde{R}_{T+1} \;-\; \tilde{q}^\pi(s_{t-1}, a_{t-1}) \mid s_{t-1}, a_{t-1}\right]\\
&= \mathrm{E}_\pi\left[\tilde{R}_{T+1} \mid s_{t-1}, a_{t-1}\right] \;-\; \mathrm{E}_\pi\left[\mathrm{E}_\pi\left[\sum_{\tau=t-1}^{T} \tilde{R}_{\tau+1} \mid s_{t-1}, a_{t-1}\right] \mid s_{t-1}, a_{t-1}\right]\\
&= \mathrm{E}_\pi\left[\tilde{R}_{T+1} \mid s_{t-1}, a_{t-1}\right] \;-\; \mathrm{E}_\pi\left[\tilde{R}_{T+1} \mid s_{t-1}, a_{t-1}\right]\\
&= 0 \;.
\end{aligned}$$

If we substitute $t-1$ by t (t one step further and m one step smaller) it follows

$$\kappa(T-t, t) \;=\; 0 \;. \tag{A137}$$

Next, we consider the case $t = T+1$, that is $\kappa(0, T)$, which is the expected correction. We will use following equality for the expected delayed reward at sequence end:

$$\tilde{q}^\pi(s_T, a_T) \;=\; \mathrm{E}_{\tilde{R}_{T+1}}\left[\tilde{R}_{T+1} \mid s_T, a_T\right] \;=\; \tilde{r}_{T+1}(s_T, a_T) \;, \tag{A138}$$

since $\tilde{q}^\pi(s_{T+1}, a_{T+1}) = 0$. For $t = T+1$ we obtain

$$\begin{aligned}
&\mathrm{E}_{R_{T+2}}\left[R_{T+2} \mid s_T, a_T\right] \;=\; \mathrm{E}_{\tilde{R}_{T+1}}\left[\tilde{R}_{T+1} \;-\; \tilde{q}^\pi(s_T, a_T) \mid s_T, a_T\right] && \text{(A139)}\\
&= \tilde{r}_{T+1}(s_T, a_T) \;-\; \tilde{r}_{T+1}(s_T, a_T) \;=\; 0 \;.
\end{aligned}$$

□

In the experiments we also use a uniform compensation where each reward has the same contribution to the compensation:

$$R_1 \;=\; h_0 \;+\; \frac{1}{T+1}\left(\tilde{R}_{T+1} - \sum_{\tau=0}^{T} h(\Delta(s_{\tau-1}, a_{\tau-1}, s_\tau, a_\tau))\right) \tag{A140}$$

$$R_{t+1} \;=\; h_t \;+\; \frac{1}{T+1}\left(\tilde{R}_{T+1} - \sum_{\tau=0}^{T} h(\Delta(s_{\tau-1}, a_{\tau-1}, s_\tau, a_\tau))\right) \;. \tag{A141}$$